

Role-Based Functionality Audit Report

Scope

Review of backend/frontend functionality, role permissions, routing, API integration, and overall system stability.

Executive Summary

The platform is partially operational. Authentication and most read/list operations function correctly, and several modules have solid structural implementation. However, the application is not yet production-ready due to backend write-operation failures, unstable migrations, incomplete frontend modules, and permission/scoping concerns.

Critical Findings

1. Backend Write Operation Failures

Many repositories still use PostgreSQL-specific `RETURNING` statements while the application runs on MySQL/MariaDB. This creates runtime failures during create/update operations across multiple modules.

Impact

- Unstable CRUD operations
- High risk of failed data persistence
- Affects admin, HR, and employee workflows

2. Migration Reliability Issues

Migration execution lacks proper rollback and failure handling. Some migration steps may fail silently while later steps continue executing.

Impact

- Partially created database schemas
- Inconsistent environments
- Deployment instability

3. Incomplete Frontend Modules

Several role-critical pages remain placeholders or partially implemented, especially:

- Time management
- Leaves
- Absences
- Holidays
- Audit logs

Impact

- Incomplete user experience
- Missing business workflows
- Reduced operational usability

4. API Contract Mismatches

Some frontend expectations do not match backend response structures.

Examples

- Logout endpoint inconsistencies
- Dashboard totals returned under unexpected pagination metadata

Impact

- Broken dashboard metrics
- Inconsistent frontend behavior
- UX instability

Role-Based Assessment

Super Admin

Functional

- Authentication and protected routing
- Employee management flows
- Document/template management
- User administration actions

Issues

- Write operations unstable
- Settings modules incomplete
- Some admin pages unfinished
- Dashboard metric inconsistencies

HR Agent

Functional

- Role-based access
- Read/list employee flows
- Dashboard data rendering

Issues

- Create/update flows unstable
- Audit module incomplete
- Leave/absence/time pages unfinished

Manager

Functional

- Authentication and dashboard access
- Basic dashboard data retrieval

Issues

- Approval actions incomplete
- Core manager modules unfinished
- Potential permission-scoping weaknesses

Employee

Functional

- Authentication and protected access
- Dashboard data retrieval

Issues

- Self-service modules incomplete
- Leave request flow unfinished
- Possible data-isolation concerns

Guest/Public

Functional

- Login access and redirects

Issues

- Forgot password not implemented
- No public-facing functionality

Recommendations

High Priority

1. Replace PostgreSQL-specific SQL syntax incompatible with MySQL/MariaDB.
2. Improve migration rollback/error handling.
3. Complete placeholder frontend modules.
4. Enforce stricter role-based data scoping and permissions.
5. Standardize frontend/backend API contracts.